From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST - III

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 240**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-B & Section–C**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 – 08)** contains 8 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (09 – 12) contains 4 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

(ii) **Section-B (01 – 02)** contains 2 Matrix Match Type questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+8 marks** for all correct answer. For each correct row **+2 marks** will be awarded. There may be one or more than one correct choice. No marks will be given for any wrong match in any question. There is no negative marking.

(iii) **Section-C (01 – 06)** contains 6 Numerical based questions with single digit integer as answer, ranging from 0 to 9 and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

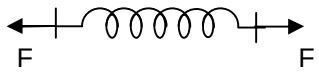
PART – I

SECTION – A

Straight Objective Type

This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. If an ideal spring is stretehed by a force F = 3N as shown then its length is observed to be 2m. If the value of F is 4N instead of 3N then which of the following cannot be it's length is second case?



(A) 2.4 m (B) 2.55 m
(C) 2.7 m (D) none of these

2. The potential energy U in joule of a perticle of mass 2 kg moving in X – Y plane follows the law U = 2x + 4y, where (x, y) are the co-ordinates (in metre) of the particle. If the particle is at rest at (4, 4) at time t = 0 then, [choose the incorrect statement]
(A) at t = $2\sqrt{2}$ sec, the paticle crosses the y-axis.
(B) at t = 2sec, the particle crosses the x-axis.
(C) The particle has kinetic energy of 40 Joule when it crosses y-axis.
(D) The kinetic energy of the particle when it crosses x-axis is 40 Joule.

3. A container of mass 'm' initially at rest and containing same mass of water in it, is being pulled vertically up by an ideal string exerting a constant force 'F' starting at t = 0. The water starts leaking out of the container with negligible relative speed. The velocity of the container at an instant t = t_0 when it becomes empty is 'v'. Then the value of t_0 is
(A) $\frac{mv}{F(\ell n2)-mg}$ (B) $\frac{mv}{F(\ell n2)+mg}$
(C) $\frac{2mv}{F(\ell n2)+2mg}$ (D) $\frac{mv}{2F(\ell n2)-mg}$

4. Three moles of an inert monoatomic gas is mixed with two moles of a diatomic gas and the mixture is heated by a process V = $\frac{k}{\sqrt{T}}$. Where V and T are volume and temeprature of the mixture and k is a constant. The molar heat capacity of this process for this mixture is
(A) $\frac{7R}{5}$ (B) $\frac{9R}{5}$
(C) $\frac{11R}{5}$ (D) none of these

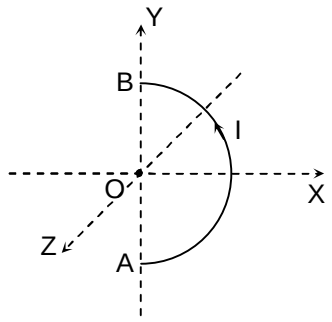
Space for rough work

5. A conducting sphere of radius R and charge Q is surrounded by a medium of resistivity 'ρ' at t = 0. Due to the medium the charge starts decreasing from it. At t = t_1 the charge on the sphere is found $\frac{Q}{4}$ then the value of t_1 is [Absolute permitivity of the vacuum is $\in_0$]

(A) $\frac{\rho \in_0 \ln 2}{2}$ (B) $\frac{\rho \in_0 \ln 2}{4}$

(C) $4\rho\in_0\ln 2$ (D) $2\rho\in_0\ln 2$

6. A conductor carrying current I is in the form of a semicircle AB of radius R and lying in xy-plane with it's centre at origin as shown in the figure. Which of the following is the magnitude of $\oint \vec{B}.\vec{d\ell}$ for the circle $3x^2 + 3z^2 = R^2$ in xz-plane due to current in curve AB?



(A) $\frac{\sqrt{3}}{2}\mu_0 I$ (B) $\left(\sqrt{3}+1\right)\frac{\mu_0 I}{2}$

(C) $\left(2-\sqrt{3}\right)\frac{\mu_0 I}{2}$ (D) zero

7. In a young's double slit interference experiment, the fringe pattern is observed on a screen placed at a distance D. The slits are separated by d (<<D) and are illuminated by coherent sources. The central maxima is formed at the geometrical centre on the screen with respect to the slits. If at a distance h from the central maximum on the screen the intensity is half of the maximum intensity then the value of λ may be

(A) $\frac{2hd}{3D}$ (B) $\frac{8hd}{19D}$

(C) $\frac{4hd}{13D}$ (D) $\frac{6hd}{11D}$

8. Momentum P of a particle is given as, P = at $+\frac{b+t^2}{c}+dt^3$, t represents time. If [X] represents dimensional expression of any physical quanity X then choose the correct.

(A) $[a] = [MLT^{-2}]$, $[b] = [M^oL^oT^2]$, $[d] = [MLT^{-3}]$ (B) $[a] = [MLT^{-2}]$, $[c] = [M^{-1}L^{-1}T^3]$, $[d] = [MLT^{-4}]$

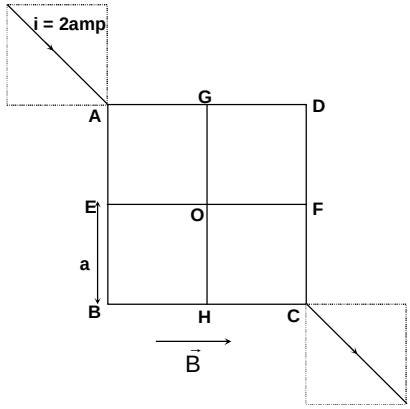
(C) $[a] = [ML^{-1}T^2]$, $[c] = [MLT^{-3}]$, $[d] = [MLT^{-4}]$ (D) $[a] = [MLT^3]$, $[b] = [M^oL^oT^{-2}]$, $[c] = [M^{-1}L^{-1}T^3]$

Space for rough work

Multiple Correct Choice Type

This section contains **4 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

9. In the figure there is a uniform conducting structure in which each small square has side a =1m. The structure is kept in uniform magnetic field B = 2 Tesla along BC. Current of 2 amperes enters the structure at A and leaves at C. Find the correct option(s).



(A) The magnetic force on the structure is $8\sqrt{2}$N

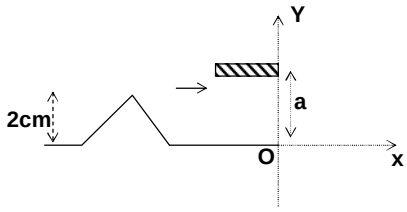
(B) The potential of point B = potential of D.

(C) Potential of O = potential of D

(D) If r = 1Ω is the resistance of small side, the equivalent resistance across AC is $\frac{3}{2}\Omega$

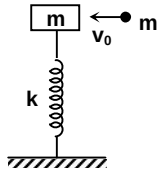
Space for rough work

10. In the figure, a wave pulse is moving on a string towards a loosely bound end and there is no loss of energy on reflection. A touch screen cell phone is kept parallel to x-axis. Cell phone is at a distance 'a' from the origin. For what value of 'a' will the cell phone screen get activated.



(A) 3 (B) 5.5
(C) 3.5 (D) 5

11. A vertical spring of spring constant k is compressed by x_0 by a block of mass m as shown in figure. Block is released from rest. Spring can be gently detached from block at any moment. At the instant of detachment, a particle of mass m moving with speed v_0 horizontally collides with the block perfectly inelastically.



(A) to have max. horizontal velocity of the block on the ground, spring can be detached at any moment.
(B) to have max. horizontal velocity of the block on the ground, spring should only be detached at it's natural length.
(C) to have max. mechanical energy of the block on the ground, spring can be detached at any moment
(D) to have max. mechanical energy of the block on the ground spring should be detached at it's natural length.

12. A cylindrical shell, a spherical shell and a solid sphere are allowed to roll on an inclined rough surface of co-efficient of friction μ and inclination θ. The correct statements are
(A) If cylindrical shell can roll on inclined plane, all other objects will also roll
(B) If all the objects have different mass and allowed to roll from same height, then K.E. of all the object will be different at the bottom of inclined plane.
(C) work done by the frictional force will be zero if object are rolling.
(D) frictional force will be equal for all the objects having same mass

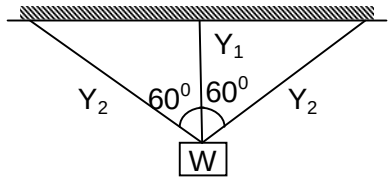
Space for rough work

SECTION – B

(Matching List Type)

This section contains **2 multiple choice questions.** Each question has matching lists. The codes for the Column(s) have choices (A), (B), (C) and (D) out of which only **ONE OR MORE** is/are correct

1. A load is supported using three wires of same cross section area as shown. Then for



	Column I		Column II
(A)	Equal tensile stress in all wires	(p)	$Y_2 = 2Y_1$
(B)	Equal tension in all wires	(q)	$Y_2 = 4Y_1$
(C)	Equal elastic potential energy in all wires	(r)	$Y_2 = 8Y_1$
(D)	Equal elastic potential energy per unit volume in all wires	(s)	$Y_2 = 16Y_1$
		(t)	Does not depend on ratio of $\frac{Y_1}{Y_2}$

2. A mono atomic ideal gas expands according to law PV^n = constant. ΔU is change in internal energy of the gas, ΔQ is heat absorbed by the gas. Match column for different values of n

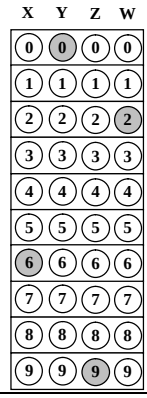
	Column I		Column II
(A)	n = 1.8	(p)	$\Delta Q > 0$
(B)	n = 1.5	(q)	$\Delta Q < 0$
(C)	n = 1.2	(r)	$\Delta U > 0$
(D)	n = 0.9	(s)	$\Delta U < 0$
		(t)	$\Delta U = 0$

Space for rough work

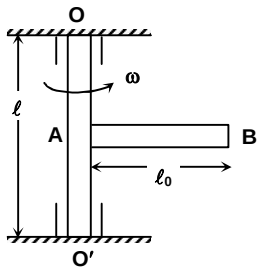
SECTION – C

Integer Answer Type

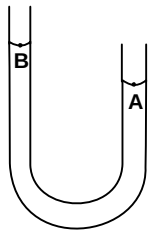
This section contains **6 questions.** The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. A horizontally uniform rod AB of mass m and length ℓ_0 rotates freely about a stationary vertical axis OO′ passing through the end A, which is located exactly midway between O and O′ and OO′ = ℓ. At what angular velocity (in units) of the rod is the horizontal component of the force acting on the lower end of the axis OO′ equal to zero. (g = 10 m/s^2, ℓ = 5m, ℓ_0 = 3m)



2. The difference in water levels in the two communicating capillary tubes of different diameter d = 1 mm and d = 1.5 mm is K × 2.38. Surface tension of water = 0.07 N/m and angle of contact between glass and water is 0°. Find the value of K.



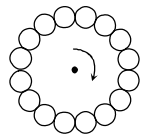
Space for rough work

3. As a result of isobaric heating by ΔT = 72 K. One mole of a certain ideal gas receives heat Q = 1.6 kJ. Find the value of adiabatic exponent ν for the gas is $8/\lambda$. Find the value of λ.

4. A coil carrying a current I = 10 mA is placed in a uniform magnetic field so that its axis coincides with the field direction. The coil consist of only one turn and made up of copper. The diameter of the wire is 0.1 mm, the radius of the coil is R = 3π cm. The value of external induction field will the coil rupture is 500K. Breaking stress of copper = 3×10^8 N/m^2. Then find the value of K.

5. A rubber ball of mass m and radius r is submerged into water of density ρ to the depth h and released. Then the height will the ball jump up above the surface of water is $\left(\rho\frac{2K}{3}\pi r^3 - m\right)\frac{h}{m}$. Find the value of K.

6. A metallic chain with a length ℓ and whose ends are joined together is fitted onto a wooden disc as shown in the figure. The disc rotates with a speed of n revolutions per second. Then the tension of the chain T if its mass = m is $Km\ell n^2$. Find the value of K.



Space for rough work

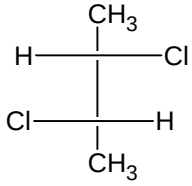
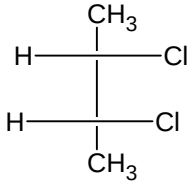
Chemistry PART – II

SECTION – A

Straight Objective Type

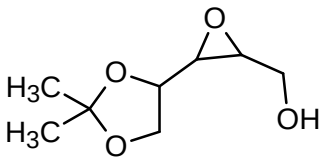
This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

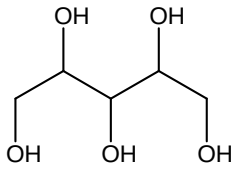
1. Heat of neutralization of HCl and NaOH is x kJ and that of CH_3COOH and NaOH is y kJ. What is heat of ionization of CH_3COOH
   (A) (x – y) kJ (B) (y – x) kJ
   (C) (x + y) kJ (D) - (x + y) kJ

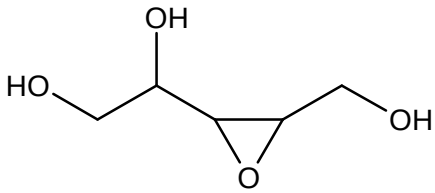
2. If the optical rotation produced by  is $+36^o$, then the optical rotation produce by  will be
   (A) -36° (B) 0°
   (C) $+36^o$ (D) Unpredictable

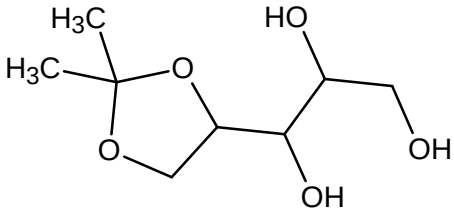
3. The IUPAC name for $[Pt(NH_3)_3(Br)(NO_2)(Cl)]Cl$ is
   (A) Triamminechlorobromonitroplatinum (IV) chloride
   (B) Triamminebromochloronitroplatinum (IV) chloride
   (C) Triamminechlorobromoplatinum (IV) chloride
   (D) Triamminechloronitrobromoplatinum (IV) chloride

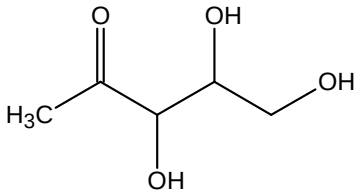
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4.  $\xrightarrow[2.\ HCl,\ H_2O]{1.\ NaOH,\ H_2O}$ A, A is

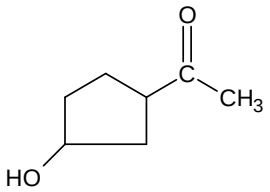
(A) 

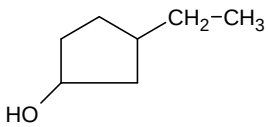
(B) 

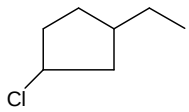
(C) 

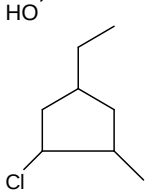
(D) 

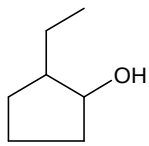
5. The solubility product constants of Ag_2CrO_4 and AgBr are 1.1×10^{-12} and 5.0×10^{-13} respectively. The ratio of molarities of their saturated solution is
(A) 216.4 (B) 91.9
(C) 121.6 (D) 87.8

6.  $\xrightarrow[HCl]{Zn(Hg)}$ (A); Identify A ?

(A) 

(B) 

(C) 

(D) 

Space for rough work

7. Element X crystallizes in a 12 coordination f.c.c lattice. On applying high temperature it changes to 8 coordination b.c.c. lattice. Find the ratio of the density of the crystal lattice before & after applying high temperature?
(A) 1:1 (B) 3:2
(C) $\sqrt{2}:\sqrt{3}$ (D) $2\left(\sqrt{2}\right)^3:\left(\sqrt{3}\right)^3$

8. When NaCl is heated in presence of sodium vapour it becomes yellow coloured due to
(A) Schottky defect (B) Frenkel defect
(C) Metal in excess defect (D) Metal in deficiency defect

Multiple Correct Choice Type

This section contains **4 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

9. Ratio between potential energy, kinetic energy and total energy of electron in hydrogen atom are
(A) $\frac{KE}{PE}=-\frac{1}{2}$ (B) $\frac{E}{PE}=\frac{1}{2}$
(C) $\frac{KE}{E}=-\frac{1}{2}$ (D) $\frac{E}{PE+KE}=-1$

10. The dissociation of ammonium carbamate is represented by the following equation:
$NH_4CO_2NH_2(s) \rightleftharpoons 2NH_3(g)+CO_2(g)$
ΔH^o for the backward reaction is positive. The equation will shift from right to the left if there is
(A) an increase in temperature
(B) a decrease in pressure
(C) an increase in concentration of ammonia
(D) addition of inert gas at constant pressure into the reaction container

11. In a solid AB having NaCl structure, A atoms occupy the corners of the cubic unit cell. If all the face centred atoms along one of the axes are removed, then the resultant stoichiometry of the solid is not:
(A) AB_2 (B) A_2B
(C) A_4B_3 (D) A_3B_4

12. Choose the correct statements:
(A) $BaCl_2+AcOH+K_2CrO_4 \longrightarrow$ Yellow ppt. (B) $BaCO_3(s)+K_2CrO_4 \longrightarrow$ Yellow ppt.
(C) $BaCO_3(s)+K_2CrO_4+AcOH \longrightarrow$ No ppt. (D) $SrCO_3(s)+K_2CrO_4+AcOH \longrightarrow$ No ppt.

Space for rough work

SECTION – B

(Matching List Type)

This section contains **2 multiple choice questions.** Each question has matching lists. The codes for the Column(s) have choices (A), (B), (C) and (D) out of which only **ONE OR MORE** is/are correct

1. Match the organic compound in Column I with the type of isomerism shown by them:

Column – I	**Column – II**
(A) $CH_3 - CH = CH - CH_2 - Cl$	(p) Chain isomerism
(B) $CH_3 - CH_2 - CH_2 - CH_2 - O - CH_3$	(q) Tautomerism
(C) $CH_3 - CH_2 - CH_2 - CH_2OH$	(r) Functional isomerism
(D) $CH_3 - COOCH_3$	(s) Position isomerism
	(t) Metamerism

2.

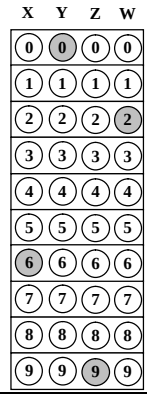
Column – I	**Column – II**
(A) $\left[Cu(NH_3)_4\right]^{2+}$	(p) Tetrahedral shape
(B) $\left[Co(ox)_3\right]^{3-}$	(q) Coloured complex and paramagnetic
(C) $[Ni(CO)_4]$	(r) Octahedral (sp^3d^2 hybridization)
(D) $\left[Ni(CN)_4\right]^{2-}$	(s) Square planar shape
	(t) Trigonal planar

Space for rough work

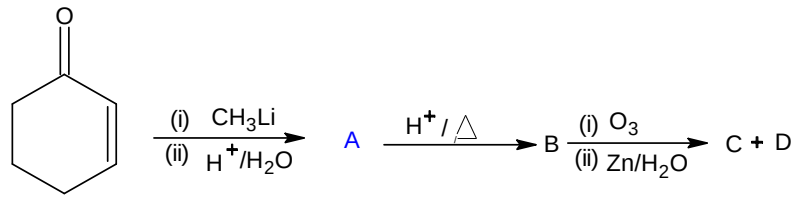
SECTION – C

Integer Answer Type

This section contains **6 questions.** The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. The first order gaseous decomposition of N_2O_4 in to NO_2 has a rate constant value $K = 4.5 \times 10^3\ s^{-1}$ at 1^oC and energy of activation 58 kJ mol^{-1}. What should be the rise in temperature in (^oC) to get the value of $K = 1.00 \times 10^4\ s^{-1}$. (log 2.2 = 0.3424)

2. 

How many moles of mild oxidizing agent are needed to oxidize the products formed?

3. How many secondary amines are possible with the formula $C_5H_{13}N$.

4. 10 ml of a solution of H_2SO_4 having pH = 5 is diluted to 10 litres by adding water, pH of final solution is....

5. At critical set of condition the value of compressibility factor is $\frac{1.5}{x}$, the value of 'x' is:

6. In analytical chemistry how many of these cations form black precipitate of sulphide:
$Ca^{2+}, Hg^{2+}, Pb^{2+}, Cu^{2+}, Zn^{2+}, As^{3+}, Cd^{2+}, Sb^{3+}, Ni^{2+}, Co^{2+}$

Space for rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. The value of the definite integral $\int_{-\infty}^{\ln 3} \{e^x\}\,dx$ equals (where {y} denote fractional part of y)

   (A) $3 + \ln 2 - 2\ln 3$ (B) $3 - \ln 3$
   (C) $2\ln 3 + \ln 2$ (D) 1

2. Let z_1 and z_2 be roots of the equation $z^2 + 8(i - 1)z + 63 - 16i = 0$ where $i^2 = -1$. The area of triangle formed by o, z_1 and z_2 (where o being origin) is equal to
   (A) 24 (B) 26
   (C) 28 (D) 30

3. Let ABC be a triangle whose vertices are A(–5, 5) and B(7, –1). If vertex c has on a circle whose director circle has equation $x^2 + y^2 = 100$, then locus of the orthocentre of triangle ABC is equal to
   (A) $x^2 + y^2 + 4x - 8x - 30 = 0$ (B) $x^2 + y^2 - 4x + 8y - 30 = 0$
   (C) $x^2 + y^2 - 4x - 8y - 30 = 0$ (D) $x^2 + y^2 + 4x + 8y - 30 = 0$

4. If the complex number $z = x + iy$ moves in argand plane such that $1 \le |x + iy| \le 2$ and $x - y \le 0$ where $x, y \in R$ determines a region with area k then value of the definite integral $\int_0^{2k/3} \{\sin x\}\,dx$ is equal to (where {k} represents fractional part of k)
   (A) 0 (B) 1
   (C) 2 (D) 3

5. A circle passes through A(0, 4) and B(8, 0) has its centre on x–axis. If point C lies on the circumference of the circle and m is the greatest area of triangle ABC, then m is equal to
   (A) $10(\sqrt{5} - 1)$ (B) $10(\sqrt{5} + 1)$
   (C) $20(\sqrt{5} + 1)$ (D) $20(\sqrt{5} - 1)$

Space for rough work

6. If $M(x_0, y_0)$ is the point on the curve $3x^2 - 4y^2 = 72$ which is nearest to the line $3x + 2y + 1 = 0$ then the value of $(x_0 + y_0)$ is equal to
(A) 3 (B) –3
(C) 9 (D) –9

7. Area enclosed by the figure described by the equation $x^4 + 1 = 2x^2 + y^2$ is
(A) 3 (B) $\frac{8}{3}$
(C) $\frac{7}{3}$ (D) $\frac{10}{3}$

8. Let $f(x) = \left[\frac{\sin x}{x}\right] + \left[\frac{2\sin 2x}{x}\right] + ... + \left[\frac{10\sin 10x}{x}\right]$ where [y] is the greatest integer function. The value of $\lim_{x\to 0} f(x)$ is equal to
(A) 55 (B) 164
(C) 165 (D) 375

Multiple Correct Choice Type

This section contains **4 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

9. A circle 'S' is described on the focal chord of the parabola $y^2 = 4x$ as diameter. If the focal chord is inclined at an angle of 45º with axis of x, then which of the following is/are true
(A) radius of the circle is 4
(B) centre of the circle is (3, 2)
(C) the line $x + 1 = 0$ touches the circle
(D) the circle $x^2 + y^2 + 2x - 6y + 3 = 0$ is orthogonal to 'S'

10. If three planes $P_1 \equiv 2x + y + z - 1 = 0$, $P_2 \equiv x - y + z - 2 = 0$ and $P_3 \equiv \alpha x - y + 3z - 5 = 0$ intersect each other at point P on XOY plane and at point Q on YOZ plane, where O is the origin then identify the correct statement(s)
(A) the value of α is 4
(B) straight line perpendicular to plane P_3 and passing through P is $\frac{x-1}{4} = \frac{y+1}{-1} = \frac{z}{3}$
(C) the length of projection of $\overrightarrow{PQ}$ on x–axis is 1
(D) centroid of the triangle OPQ is $\left(\frac{1}{3}, -\frac{1}{2}, \frac{1}{2}\right)$

Space for rough work

11. PQ is a double ordinate of the parabola $y^2 = 4ax$. If the normal at P intersect the line passing through Q and parallel to x–axis at G then locus of G is a parabola with
(A) length of latus rectum equal to 4a (B) vertex at (4a, 0)
(C) directrix as the line x – 3a = 0 (D) focus at (5a, 0)

12. If α is a root of the equation $4x^2 + 2x - 1 = 0$ then the other root must be
(A) $3\alpha^3 - 4\alpha$ (B) $4\alpha^3 - 3\alpha$
(C) $2\alpha^2 - 1$ (D) all above

SECTION – B

(Matching List Type)

This section contains **2 multiple choice questions.** Each question has matching lists. The codes for the Column(s) have choices (A), (B), (C) and (D) out of which only **ONE OR MORE** is/are correct

1. Match the following Column–I with Column–II

	Column – I		Column – II
(A)	If $\lim_{x\to 3} \frac{f(x)-3}{x-2} = 1$ then $\lim_{x\to 3} f(x)$ is equal to	(p)	15
(B)	The number of order pairs of integers (x, y) satisfying the equation $x^2 + 6x + y^2 = 4$ is	(q)	11
(C)	Consider the graphs of y = sin x, y = cos x, y = tan x, y = cot x, y = sec x and y = cosec x Let $R = \left\{(x, y) \middle\| 0 \le x \le \frac{\pi}{2},\ 0 \le y \le 100\right\}$ Number of points of R which be on atleast two of the graph, is	(r)	8
(D)	Suppose that $2 - \sqrt{95}$ is a root of $x^2 + ax + b = 0$ where b is negative real number and a is a integer. The largest possible value of a, is	(s)	7
		(t)	4

Space for rough work

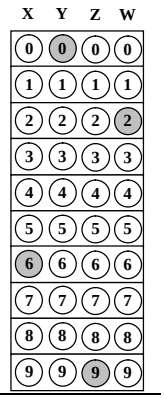
2. Match the following Column–I with Column–II

Column – I		Column – II	
(A)	Number of roots of the equation $\log_{\frac{6x-x^2}{11}}(-\cos x-\cos 3x)=\log_{\frac{6x-x^2}{11}}(-\cos 2x)$ in [0, 2π] is equal to	(p)	1
(B)	If a_1 is the value of a for which $f(x)=x^2+\frac{a}{x}$ has a local minima at x = 2 and a_2 is the value of a for which f(x) has a point of inflection at x = 1, then $\left(\frac{a_1+a_2}{3}\right)$ is equal to	(q)	2
(C)	Number of points where the function f(x) = $\|\|x\|^2-2\|x\|-3\|$ is not differentiable is equal to	(r)	3
(D)	If $\lim_{x\to 0}\int_0^x \frac{t^2dt}{(x-\sin x)\sqrt{a+t}}=1$ then the value of a is equal to	(s)	4
		(t)	5

SECTION – C

Integer Answer Type

This section contains **6 questions.** The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown.



1. The number of five digit numbers that can be formed using its digits 1, 2, 3, 4, 5, 6, 7, 8, 9 in which one digit appears once and two digits appear twice (e.g. 44533) is k then the value of $\frac{k}{1890}$ is equal to _____

Space for rough work

2. Let k be the coefficient of x^{18} in the polynomial $f(x) = (1+x)^{20} + x(1+x)^{19} + x^2(1+x)^{18} + ... + x^{18}(1+x)^2$ then the value of $\frac{k}{190}$ is equal to _____

3. Let P be the sum of integral values of b in the interval [1, 20] for which the equation $4^x + (1 - b)2^{x+1} + b = 0$ has roots of apposite sign, then the value of $\frac{p}{51}$ is equal to _____

4. Let O be the interior point of ΔABC such that $2\overrightarrow{OA} + 3\overrightarrow{OB} + 6\overrightarrow{OC} = 0$ where O is origin. If $\frac{\text{Area of } (\Delta ABC)}{\text{Area of } (\Delta AOB)} = \frac{m}{n}$, where m and n are relatively prime, then (m – n) is equal to _____

5. If the quadratic equation $x^2 + (2 - \tan\theta)x - (1 + \tan\theta) = 0$ has two integral roots, then sum of all possible values of θ in the interval $(0, 2\pi)$ is $k\pi$, then the value of k is equal to _____

6. Bag 'A' contains 9 green and 1 red ball, Bag 'B' contains 10 balls all of them being green, 9 balls are drawn randomly from Bag 'A' and put into bag 'B', and then 9 balls are randomly transferred from bag 'B' to bag 'A'. If $\frac{p}{q}$ (in its lowest form) is the chance that the red ball is still there in the bag 'A' then the value of (q – p) is equal to _____

Space for rough work